# Minor- III: Probability and Statistics

Center For Integrated Studies
University of Hyderabad

November 03, 2026

**Duration:** 60 minutes
**Maximum Score:** 20 points

**Instructor: Dharmendra Kumar**
**Course Code: MM 212**

**Instructions:** Answers without justification will receive a score of zero.

1. If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{1}{2} & 0 \le b < 1 \\ \frac{3}{5} & 1 \le b < 2 \\ \frac{4}{5} & 2 \le b < 3 \\ \frac{9}{10} & 3 \le b < 3.5 \\ 1 & b \ge 3.5. \end{cases}$$

   Find $P(X = 2)$ and $P\left(\frac{1}{2} < X < \frac{3}{2}\right)$. [ 2 + 2]

2. Suppose that a biased coin that lands on heads with probability $p$ is flipped 10 times. Given that a total of 6 heads results, find the conditional probability that the first 3 outcomes are:

   (*a*) $H, T, T$ (means the first flip results in heads, the second in tails, and the third in tails ) [4]

   (*b*) $T, H, T$. [4]

3. Find $\mathrm{Var}(X)$, where $X$ is normal random variable with parameters $\mu$ and $\sigma^2$. [4]

4. Let $\Phi$ be distribution function of standard normal RVs.
   True / False: $\Phi(-x) = \Phi(x)$. [4]

All the best !